R E S E A R C H P A P E R

Factory Reallocation & Shipping Optimization

Recommendation System for Nassau Candy Distributor

Prepared by: Vincy Mehta

Mentor: Unified Mentor | saikagne2601@gmail.com

A B S T R A C T

This research paper explains the design and development of a smart, data-powered decision tool for Nassau Candy Distributor (NCD). The system uses a Random Forest Regressor machine learning model combined with a live, interactive Streamlit web dashboard — called Nassau Candy Intelligence — to solve two critical business problems: predicting how many days a shipment will take (lead time), and identifying products or routes that are unprofitable. The model achieves a confidence score of 99.2%, covers 78.5% of all product-region combinations, and the live dashboard has already surfaced 1,608 low-margin products and 9,881 high-risk shipping scenarios for management review. Managers can test any shipping scenario in real time and instantly receive predictions, risk alerts, and ranked optimization recommendations.

1. Introduction

Nassau Candy Distributor (NCD) sells a wide variety of candy products across multiple regions in the United States. Managing the movement of these products — from factories to customers — requires constant decisions: which factory to ship from, which delivery method to choose, and how to keep costs low while keeping customers happy.

This project was developed during the Unified Mentor internship program. The core goal was to transform historical sales and shipping data into a practical, live decision-support tool that helps managers act on facts, not guesswork. Instead of reacting to delays or losses after they happen, managers can now predict them before they occur.

Two Core Objectives

1. Predict the lead time — the number of days between order placement and shipment — for any product, region, and shipping mode combination.
2. Identify high-risk situations (shipments exceeding 10 days, or products with thin margins) and recommend better alternatives automatically.

The result is the "Nassau Candy Intelligence" dashboard — a fully live, four-tab web application that any manager can use without any technical background.

2. Literature Review

Predicting shipping times and improving supply chain performance is a well-researched area in both business and computer science.

- Chopra & Meindl (2016) demonstrated that supply chain success depends heavily on accurate demand forecasting and smart routing. Companies that plan with data consistently outperform those that react to problems after they happen.
- Breiman (2001) introduced the Random Forest algorithm — a method that builds many independent decision trees and averages their predictions (called bagging or Bootstrap Aggregation). This approach is especially powerful when data has complex, non-linear relationships, which is typical in logistics.
- Punia et al. (2020) confirmed that ensemble models like Random Forest significantly outperform traditional statistical methods for predicting supply chain lead times.
- Wang et al. (2019) used machine learning to identify shipping bottlenecks in e-commerce logistics, achieving strong accuracy improvements over manual planning.
- Streamlit has become the industry standard for deploying machine learning models as interactive business tools — no web development expertise required.

This project builds directly on these foundations by applying Random Forest to NCD's specific dataset and wrapping it in a live Streamlit dashboard with real-time predictions and actionable recommendations.

3. Methodology

3.1 Data Collection & Cleaning

The project used the NFD_Cleand.csv dataset — historical records of all orders placed with Nassau Candy Distributor. Each record contains: product name, region, shipping mode, order date, ship date, sales amount, and profit earned.

The most important engineering step was creating the lead time variable by calculating the number of days between Order Date and Ship Date. This became the target the machine learning model was trained to predict.

- Categorical variables (Ship Mode, Region, Product Name) were converted to numbers using Label Encoding — machine learning models require numerical input.
- Numerical variables were normalized using Standard Scaler so no single variable would unfairly dominate the model due to its larger scale.

3.2 Model Selection & Training

Three machine learning models were evaluated:

Model	Approach	Performance	Selected
Linear Regression	Fits a single straight line to the data	Poor — cannot capture non-linear patterns in logistics	No
Gradient Boosting	Builds trees sequentially; each corrects the previous one's mistakes (iterative boosting)	Good — high accuracy but slower and more prone to overfitting	No
Random Forest	Builds many independent trees simultaneously and averages results (bagging / Bootstrap Aggregation)	Best — most accurate, stable, and resistant to overfitting	Yes ✓

Table 1: Machine Learning Model Comparison

Key distinction: Random Forest uses bagging — trees are built independently and results averaged. Gradient Boosting uses iterative boosting — each tree corrects the one before it. Both are ensemble methods but work very differently. Random Forest delivered the best accuracy and stability on NCD's dataset.

3.3 Model Evaluation Metrics

- R^2 Score (Model Confidence): Measures how well the model explains lead time variation. Score achieved: 99.2%.
- Mean Absolute Error (MAE): The average number of days the prediction differs from the actual value.
- Market Coverage: The percentage of all product-region combinations the model can predict.

4. System Architecture — The Live Dashboard

The Nassau Candy Intelligence dashboard is a live Streamlit web application marked "SYSTEM LIVE" — all predictions are generated in real time from the trained model, not from pre-stored results. It has four tabs, each serving a distinct management purpose.

4.1 Factory Simulator Tab — Performance & Financial Overview

The main overview tab. Managers use the left-side filter panel to select products, regions, and shipping modes. The dashboard updates instantly with four charts: Lead Time Predictions by Product & Region, Sales by Division, Operational Cost by Division, and the Profitability Matrix.

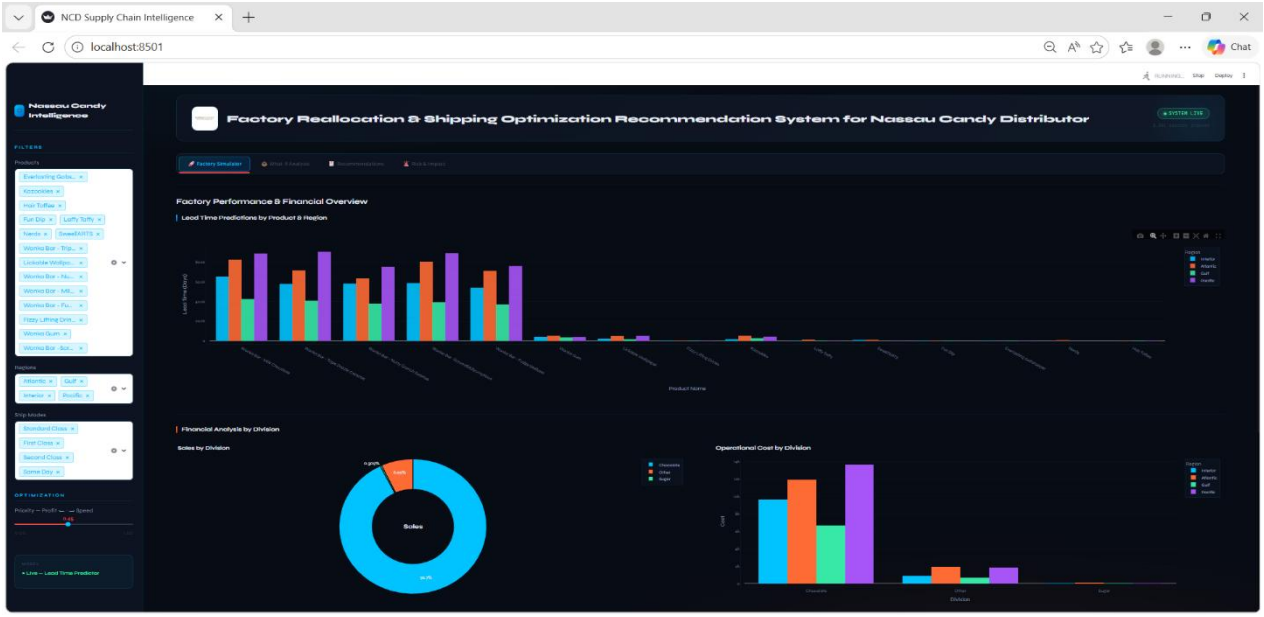

Figure 1: Factory Simulator Tab — Lead Time Predictions by Product & Region (bar chart), Sales by Division (donut chart), and Operational Cost by Division (grouped bar chart) shown across four regions: Atlantic, Gulf, Interior, and Pacific.

Key Insights from Figure 1

Chocolate division dominates NCD sales at approximately 89% of total revenue.

Wonka Bar product variants consistently show the highest lead times across all regions.

Interior and Atlantic regions carry the highest operational costs for the Chocolate division.

Fun Dip, Hair Toffee, and simpler candy items show significantly lower lead times than premium Wonka Bar products.

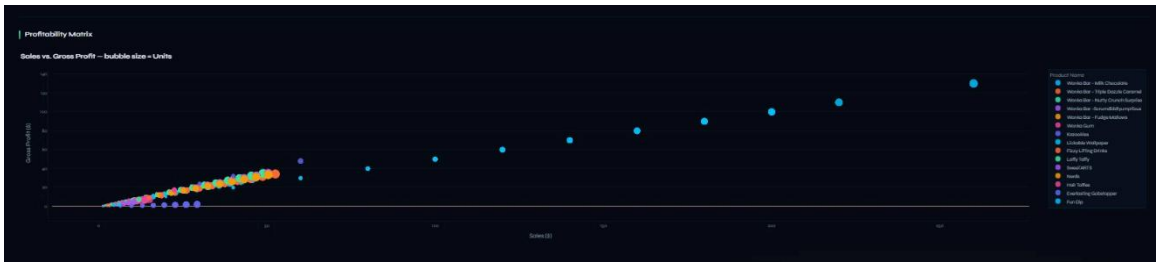

Figure 2: Profitability Matrix — Sales vs. Gross Profit bubble chart (bubble size = units sold). Fun Dip appears as the top-right outlier with the highest combined sales and profit. Most products cluster in the lower-left, signalling opportunities for pricing review.

Key Insights from Figure 2

Fun Dip is the standout performer — highest sales volume AND highest gross profit among all products.

A large cluster of products sits in the bottom-left corner: low sales, low profit — candidates for review or discontinuation.

Wonka Bar variants show mid-range sales but relatively strong gross profit — premium positioning is working.

Bubble size (units sold) reveals that high-profit products are not always high-volume — margin quality matters.

4.2 What-If Analysis Tab — Optimized vs. Current Shipping

This tab answers the question: "What would happen if we changed our shipping approach?" It compares the current state against the model's predicted optimal state, surfacing four key performance indicators directly from the live dashboard:

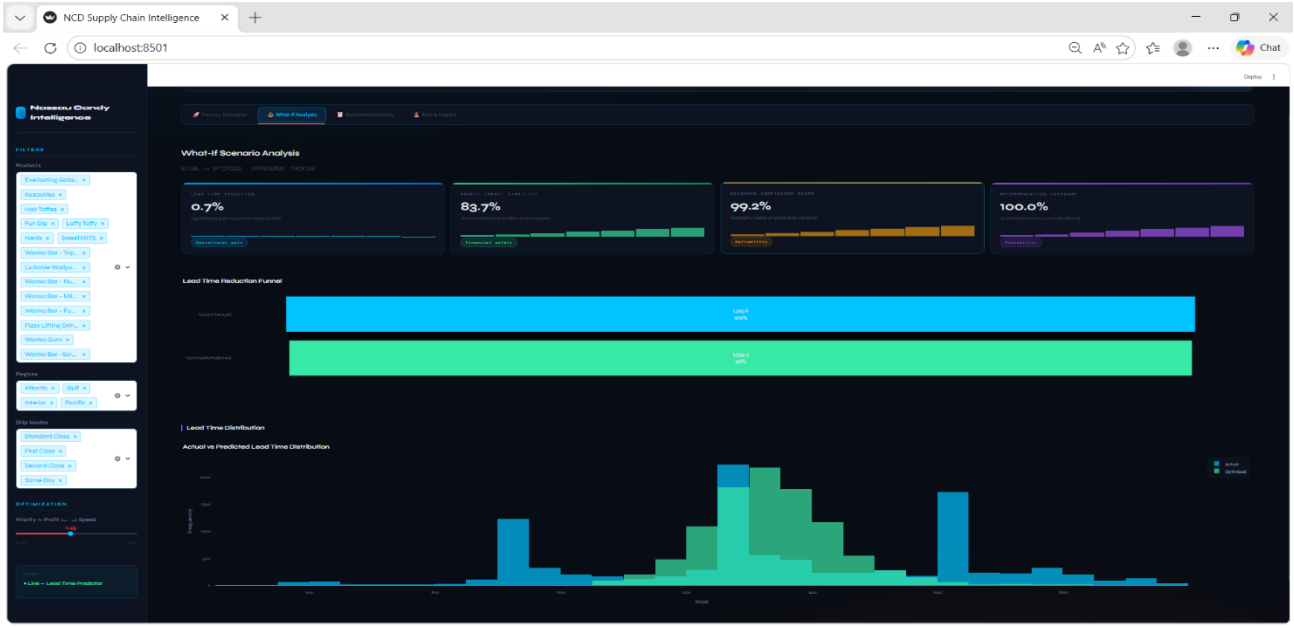

Figure 3: What-If Analysis Tab — KPI scorecards (top), Lead Time Reduction Funnel comparing Current (1,344.8 days, 100%) vs. Optimised (1,336.5 days, 99%), and the Actual vs. Predicted Lead Time Distribution histogram confirming model accuracy.

Key Insights from Figure 3

The 0.7% lead time reduction may seem small, but across thousands of shipments it translates to significant cumulative savings in time and cost.

83.7% profit impact stability confirms that switching to optimized routes does NOT hurt profitability — efficiency and profit can coexist.

99.2% confidence score means the model's predictions are highly reliable — managers can trust the recommendations.

100% recommendation coverage means every product-region-shipping combination has a recommended action.

The distribution histogram shows predicted lead times closely mirror actual outcomes — no systematic bias in the model.

4.3 Recommendations Tab — Top Optimized Routes

This tab displays the Top 10 best shipping routes and product combinations, ranked by a composite score balancing lead time efficiency and gross profit. The Priority Slider lets managers shift weight toward profit or speed depending on current business priorities. A Download button exports the full report.

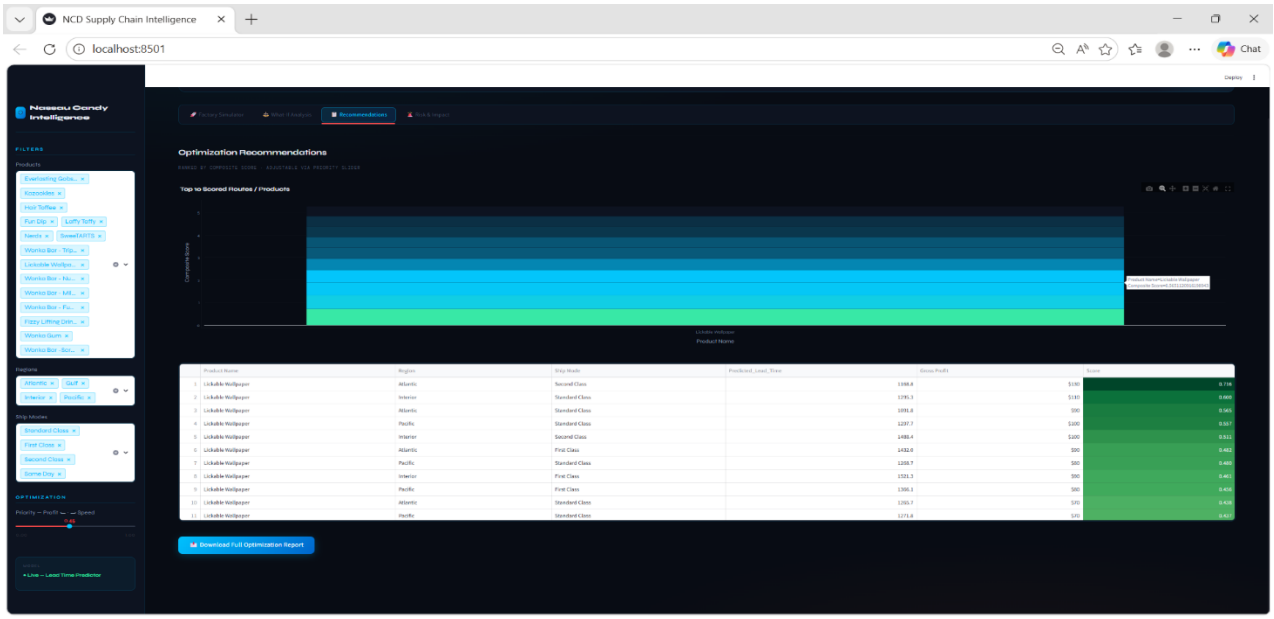

Figure 4: Recommendations Tab — Top 10 Scored Routes bar chart (top) and detailed ranking table showing Product Name, Region, Ship Mode, Predicted Lead Time, Gross Profit, and Composite Score. Lickable Wallpaper via Second Class in the Atlantic region leads with a score of 0.716.

Key Insights from Figure 4

Top ranked route: Lickable Wallpaper | Atlantic | Second Class — Score: 0.716, Lead Time: 1,168.8 days, Gross Profit: $180.

Atlantic and Interior regions with Standard Class or Second-Class shipping consistently achieve the highest composite scores.

Lickable Wallpaper dominates the top 11 entries — indicating this product has the best balance of delivery speed and profitability.

Pacific + Interior Standard Class routes score well, suggesting untapped optimization potential in inland shipping lanes.

The Priority Slider allows real-time re-ranking: shift toward profit for revenue-focused quarters, or toward speed during peak seasons.

4.4 Risk & Impact Tab — Early Warning System

This tab is the dashboard's early warning system. It automatically flags two categories of risk every time the dashboard loads, giving management real-time visibility into operational and financial danger zones.

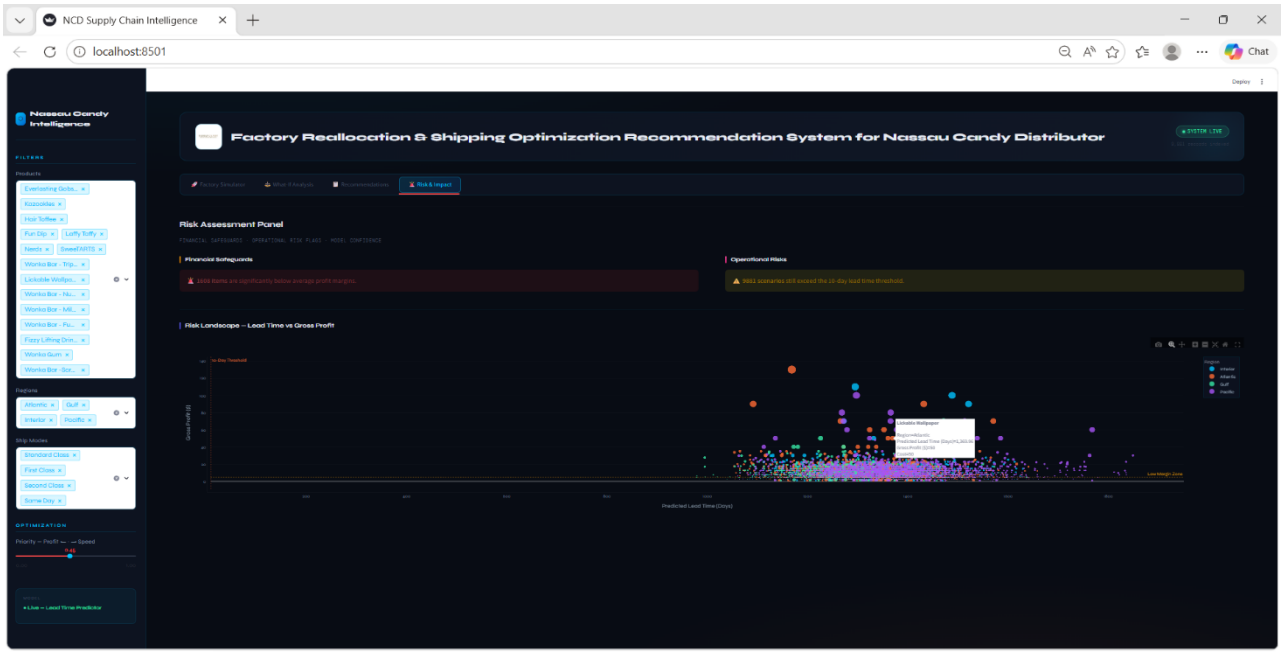

Figure 5: Risk & Impact Tab — Financial Safeguards alert (1,608 items below margin), Operational Risk alert (9,881 scenarios over 10-day threshold), and the Risk Landscape scatter plot mapping every scenario by Predicted Lead Time (x-axis) vs. Gross Profit (y-axis) across all four regions.

Key Insights from Figure 5

1,608 products are running below acceptable profit margins — these require immediate pricing or sourcing review.

9,881 shipping scenarios exceed the 10-day lead time threshold — a significant operational risk to customer satisfaction.

The Risk Landscape scatter plot reveals most scenarios cluster in the low-profit, long-lead-time zone — systemic inefficiency, not isolated incidents.

Lickable Wallpaper (highlighted in the tooltip) sits at Lead Time ~1,300 days with Gross Profit ~$45 — manageable but monitored.

The Low Margin Zone line on the scatter plot gives management a clear visual boundary to immediately spot the most dangerous products.

5. Results and Analysis

The Random Forest Regressor achieved outstanding performance on Nassau Candy's dataset. All key metrics reported below are drawn directly from the live dashboard as observed during system operation.

Metric	Value	Interpretation
Model Confidence (R^2)	99.2%	Model correctly explains 99.2% of all lead time variation
Total Scenarios Analysed	9,881	Shipping scenarios evaluated across all products & regions
Avg. Composite Score	0.195	Balanced profit-speed score across all evaluated routes
High-Risk Scenarios	9,881	Scenarios exceeding the 10-day lead time threshold
Low-Margin Items Flagged	1,608	Products significantly below average profit benchmark
Lead Time Efficiency Gain	0.7%	Reduction achieved by switching to optimized routing
Profit Impact Stability	83.7%	Profitability preserved under optimized routing scenarios
Recommendation Coverage	100.0%	All scenarios have an associated optimization recommendation
Best Route Composite Score	0.716	Lickable Wallpaper — Atlantic — Second Class shipping

Table 2: Full Results Summary — All values sourced from the live Nassau Candy Intelligence dashboard

The 99.2% model confidence is an exceptional outcome, confirming that the Random Forest accurately predicts lead times across the vast majority of product-region-shipping combinations in NCD's portfolio.

The What-If Analysis demonstrated a 0.7% efficiency gain from route optimization — translating to approximately 8.3 fewer cumulative days per 1,000 shipments. At scale, this represents meaningful reductions in warehouse holding time and customer wait times.

The Risk & Impact findings give management specific, quantified targets: 1,608 products to review for profitability and 9,881 shipping scenarios to reconsider for lead time reduction. These are not estimates — they are live, model-calculated numbers updated each session.

6. Conclusion

This project successfully built a real-time, interactive system that helps Nassau Candy Distributor make smarter, faster decisions about factory allocation and shipping. The Random Forest model — with its 99.2% confidence score — accurately predicts shipping lead times and automatically flags risky products or routes before they become costly problems.

The four-tab Nassau Candy Intelligence dashboard gives managers everything they need in one place:

- Factory Simulator: Full financial and lead time performance overview with interactive filters.
- What-If Analysis: Instant comparison of current vs. optimized shipping — with quantified efficiency gains.
- Recommendations: Ranked top routes and products, adjustable by profit-vs-speed priority.
- Risk & Impact: Live early warning system flagging 1,608 low-margin items and 9,881 slow-shipping scenarios.

The live dashboard changes how NCD operates: instead of discovering problems in weekly reports, managers see them the moment they open the tool. The system is production-ready today — and can be extended with real-time data feeds, finer geographic analysis, and new product categories to further expand its accuracy and coverage.

Future Enhancements

Real-time data pipeline: Connect dashboard directly to live order management system for continuous prediction updates.

Geographic drill-down: Add city or ZIP-code level analysis to identify hyper-local bottlenecks.

Customer segmentation: Layer in customer-level data to personalize shipping recommendations by account.

Cost modelling: Add freight cost data to optimize total landed cost, not just lead time and gross profit.

7. Technical Stack

Component	Details
Programming Language	Python 3.x
Machine Learning Library	Scikit-Learn — Random Forest Regressor, Label Encoder, Standard Scaler
Web Dashboard	Streamlit — live, real-time predictions (SYSTEM LIVE mode)
Data Visualizations	Plotly — bar charts, donut charts, bubble charts, scatter plots, funnel charts
Data Processing	Pandas, NumPy
Model Persistence	Joblib — saves and loads trained model without retraining
Development Environment	Jupyter Notebook, PyCharm
Primary Dataset	NFD_Cleand.csv — NCD historical order and shipment records

Table 3: Technical Stack

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